Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 1 of 9 · Question 6 part a.ii · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question Number					
6	(a)	(i)	phagocytosis;		1
		(ii)	• enclosed bacterium fuses with lysosome; • enzymes break down /digests bacterium; • remains of bacterium removed from white blood cell;		3
		(iii)	• label/identify/complementary to bacterium/antigens presented; • so response quicker if new infection/better immune response;		2
	(b)	(i)	any three from • prick skin to obtain blood; • use a sterile needle/pin • smear/spread blood on slide; • ensure smear is thin; • add stain; • allow to dry;		3
		(ii)	any four from • red blood cells would be seen as circle/not in 3D; • white blood cells would be larger than red/rbc too large; • wouldn't see platelets; • because too small for light microscope;		4
		(iii)	• no risk of infection; • e.g. AIDS;	R contaminate	2

Total 15

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 1 of 9 · Question 6 part a.ii · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 2 of 9 · Question 2 part a.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

2 (a) (i)			1
(ii)	In the following order only: phagocytes; enzymes;		 1 1
(iii)	produce antibodies/release antitoxins;		1
(b)	<table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table>	One mark for each correct row	3
(c)	**D** (a single-stranded helix containing the bases AUGC); A is incorrect as RNA is not a double-stranded helix B is incorrect as RNA is not a double-stranded helix C is incorrect as RNA does not contain the base T		1

Total question 2 = 8 marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 2 of 9 · Question 2 part a.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 3 of 9 · Question 2 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a) (i)			1
(ii)	In the following order only: phagocytes; enzymes;		 1 1
(iii)	produce antibodies/release antitoxins;		1
(b)	<table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table>	One mark for each correct row	3
(c)	**D** (a single-stranded helix containing the bases AUGC); A is incorrect as RNA is not a double-stranded helix B is incorrect as RNA is not a double-stranded helix C is incorrect as RNA does not contain the base T		1

Total question 2 = 8 marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 3 of 9 · Question 2 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 4 of 9 · Question 2 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a) (i)			1
(ii)	In the following order only: phagocytes; enzymes;		 1 1
(iii)	produce antibodies/release antitoxins;		1
(b)	<table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table>	One mark for each correct row	3
(c)	**D** (a single-stranded helix containing the bases AUGC); A is incorrect as RNA is not a double-stranded helix B is incorrect as RNA is not a double-stranded helix C is incorrect as RNA does not contain the base T		1

Total question 2 = 8 marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 4 of 9 · Question 2 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 5 of 9 · Question 6 part a.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
6	(a)	(i)	phagocytosis;		1
		(ii)	• enclosed bacterium fuses with lysosome; • enzymes break down /digests bacterium; • remains of bacterium removed from white blood cell;		3
		(iii)	• label/identify/complementary to bacterium/antigens presented; • so response quicker if new infection/better immune response;		2
	(b)	(i)	any three from • prick skin to obtain blood; • use a sterile needle/pin • smear/spread blood on slide; • ensure smear is thin; • add stain; • allow to dry;		3
		(ii)	any four from • red blood cells would be seen as circle/not in 3D; • white blood cells would be larger than red/rbc too large; • wouldn't see platelets; • because too small for light microscope;		4
		(iii)	• no risk of infection; • e.g. AIDS;	R contaminate	2

Total 15

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 5 of 9 · Question 6 part a.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 6 of 9 · Question 8 part a.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally;	accept through breast milk/placenta	
		• artificial passive immunity are when antibodies are injected into the body;	accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 7 of 9 · Question 8 part a.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally; • artificial passive immunity are when antibodies are injected into the body;	accept through breast milk/placenta accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 8 of 9 · Question 8 part a.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally;	accept through breast milk/placenta	
		• artificial passive immunity are when antibodies are injected into the body;	accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 9 of 9 · Question 4 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• 5 000 000 / 100 000 = 50; • 50 × 445 = 22 250; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • 445 / 185 ; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 9 of 9 · Question 4 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks